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**Effect of hydrogen enrichment on the ignition delay, laminar
flame speed and NO_x emissions of methane–air mixtures under
aero gas turbine combustor conditions**

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Abstract

This work quantifies how blending hydrogen into methane changes three combustion quantities of direct relevance to an aero gas turbine combustor: the auto-ignition delay time, the laminar burning velocity and the NO_x emission index. All calculations are performed with the Cantera library and the GRI-Mech 3.0 reaction mechanism. The hydrogen content is specified as a fraction of the chemical energy of the fuel, which is the physically meaningful variable for a fuel-substitution (decarbonisation) scenario. Parametric studies are carried out over the initial temperature, pressure, equivalence ratio and hydrogen fraction. The results show that hydrogen markedly shortens the ignition delay and strongly raises the flame speed, but that the accompanying rise in flame temperature produces a steep, near-exponential increase in thermal NO_x . The choice of GRI-Mech 3.0 across the whole blend range is validated against a dedicated detailed H_2/O_2 mechanism, with agreement better than 3.3% for the flame speed and better than 0.1% for the ignition delay of pure hydrogen.

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1 Introduction

Hydrogen is one of the leading candidates for the decarbonisation of gas turbine combustion. Because it contains no carbon, replacing part of a hydrocarbon fuel with hydrogen reduces the carbon dioxide produced per unit of energy released, and in the limit of pure hydrogen eliminates it entirely. For aero gas turbines this is particularly attractive, but it raises a practical question: how does the substitution change the combustion behaviour the engine was designed around?

Three quantities capture most of what matters for combustor design and operation. The *ignition delay* governs relight at altitude and the stability of the reaction zone. The *laminar burning velocity* sets the characteristic flame scale and is closely tied to flashback and blow-off limits. The *NO_x emission* determines whether the engine meets increasingly strict emission regulations. Hydrogen affects all three, and not always in the same direction, so a quantitative picture of the trade-offs is needed.

The aim of this project is to compute these three quantities for methane–hydrogen blends across the full range from pure methane to pure hydrogen, under conditions representative of an aero gas turbine combustor, and to identify the dominant trends and the trade-offs they imply. All calculations are performed with Cantera, an open-source chemical-kinetics library, using the GRI-Mech 3.0 reaction mechanism.

2 State of the art

Methane is the principal component of natural gas and a well-characterised gas turbine fuel, and its combustion chemistry is described accurately by established mechanisms such as GRI-Mech 3.0, which was optimised against a large body of natural-gas combustion data and includes the nitrogen sub-chemistry required for NO_x prediction. Hydrogen has a far higher reactivity, a much larger laminar burning velocity and a higher adiabatic flame temperature than methane at the same equivalence ratio, all of which make hydrogen-enriched flames more prone to thermal NO_x formation while also improving ignition and stability.

The dominant NO_x pathway in lean, high-temperature gas turbine combustion is the thermal (Zeldovich) route, in which atomic oxygen and nitrogen react at high temperature to form nitric oxide. Because the rate of this route depends exponentially on temperature, even a moderate increase in flame temperature translates into a large increase in NO_x. Lean operation, which lowers the flame temperature, is therefore the standard lever for keeping NO_x within limits, and it is the central tension when hydrogen, which raises the flame temperature, is introduced.

This study treats the methane–hydrogen system parametrically and uses a single, internally consistent mechanism over the whole blend range so that the trends are not confounded by mechanism changes. The validity of that choice at the pure-hydrogen end is examined explicitly in Section 4.4.

3 Model description

All three quantities are computed with Cantera and the GRI-Mech 3.0 mechanism (53 species, 325 reactions). The code is organised into three independent modules, one per quantity, that share a common module defining the fuel blend and the mixture state. Air is modelled as O₂:N₂ = 1:3.76 by mole.

3.1 Fuel blend definition

The hydrogen content of the blend is specified as a fraction x_E of the *chemical energy* supplied by hydrogen, not as a mole fraction. This is the meaningful variable for the substitution question “what happens if a fraction x_E of the fuel energy is provided by hydrogen?”. If n_{H_2} and n_{CH_4} are the molar amounts of the two fuels and LHV denotes the molar lower heating value, then

$$x_E = \frac{n_{\text{H}_2} \text{LHV}_{\text{H}_2}}{n_{\text{H}_2} \text{LHV}_{\text{H}_2} + n_{\text{CH}_4} \text{LHV}_{\text{CH}_4}}, \quad (1)$$

with $\text{LHV}_{\text{CH}_4} = 802.3 \text{ MJ/kmol}$ and $\text{LHV}_{\text{H}_2} = 241.8 \text{ MJ/kmol}$. Inverting this relation gives the mole fractions used to set the Cantera mixture. Because hydrogen carries much less energy per mole than methane, a given energy fraction corresponds to a substantially larger mole fraction: an energy share of 30 % hydrogen, for instance, is close to 59 % of the moles. Table 1 lists the mapping.

Table 1: Mole fractions of the CH_4/H_2 blend corresponding to a given hydrogen energy fraction x_E .

x_E [%]	X_{CH_4}	X_{H_2}
0	1.000	0.000
10	0.731	0.269
30	0.413	0.587
50	0.232	0.768
70	0.114	0.886
90	0.032	0.968
100	0.000	1.000

3.2 Ignition delay

The auto-ignition delay is computed in an adiabatic, constant-volume reactor (Cantera `IdealGasReactor`), which corresponds to the classical shock-tube definition. The homogeneous mixture is integrated in time and the ignition delay τ_{ign} is taken as the instant of maximum temperature-rise rate, $dT/dt|_{\text{max}}$. This definition is robust and mechanism-independent. The delay is studied as a function of the initial temperature (900–1400 K), the initial pressure (1–40 bar) and the hydrogen energy fraction, at a lean equivalence ratio of $\phi = 0.7$.

3.3 Laminar burning velocity

The laminar burning velocity is obtained from a one-dimensional, freely-propagating, adiabatic premixed flame (Cantera `FreeFlame`) with mixture-averaged transport and adaptive grid refinement. S_L is the unburned-gas inlet velocity of the converged solution. The flames are solved at an unburned temperature of 450 K and a pressure of 5 bar, as a function of equivalence ratio (0.5–1.4) and hydrogen energy fraction. Successive points in a sweep are solved by continuation, reusing the previous converged profile as the initial guess.

3.4 NO_x emissions and flame temperature

The NO_x calculation proceeds in two steps. First the fresh charge is brought to its constant-pressure adiabatic equilibrium state, which fixes the burned-gas temperature T_{ad} and the major-species and radical pool. The nitrogen species are then reset to zero and the hot products are

integrated kinetically at constant pressure for a residence time of 5 ms, representative of a gas turbine primary zone. This yields the NO formed within the available residence time, which is more realistic than the chemical-equilibrium value because thermal NO formation is rate-limited. The result is reported as an emission index E_{INOx} , in grams of NO_2 -equivalent per kilogram of fuel, with all NO converted to an equivalent mass of NO_2 . The study is performed at 700 K and 20 bar as a function of equivalence ratio (0.5–1.3) and hydrogen fraction.

4 Results

In every figure the curves are labelled by the hydrogen energy fraction, from 0 % (pure methane) to 100 % (pure hydrogen).

4.1 Ignition delay

Figure 1 shows the ignition delay as a function of initial temperature in Arrhenius form. For every blend the data fall on a near-straight line in $\log \tau_{\text{ign}}$ versus $1/T_0$, confirming Arrhenius-type behaviour. For pure methane the delay falls from about 294 ms at 900 K to a fraction of a millisecond at 1400 K; hydrogen-rich blends ignite far faster and the separation between the curves widens with temperature.

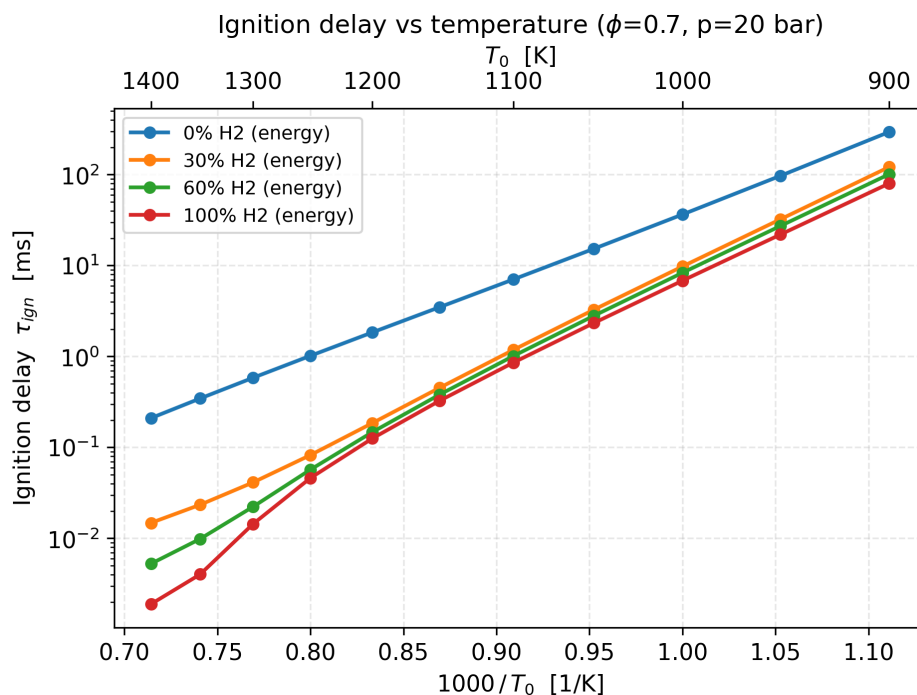


Figure 1: Ignition delay versus initial temperature for different hydrogen energy fractions ($\phi = 0.7$, $p = 20$ bar).

The effect of pressure is shown in Figure 2. Increasing the pressure shortens the ignition delay for all blends, as expected from the higher collision frequency. For methane the dependence is close to a single power law over the whole range, whereas the hydrogen-rich blends show the non-monotonic behaviour characteristic of the hydrogen explosion limits at low pressure.

Figure 3 isolates the effect of hydrogen at fixed temperature and pressure. The ignition delay drops steeply with the first few per cent of hydrogen and then more gently: at $T_0 = 1100$ K and

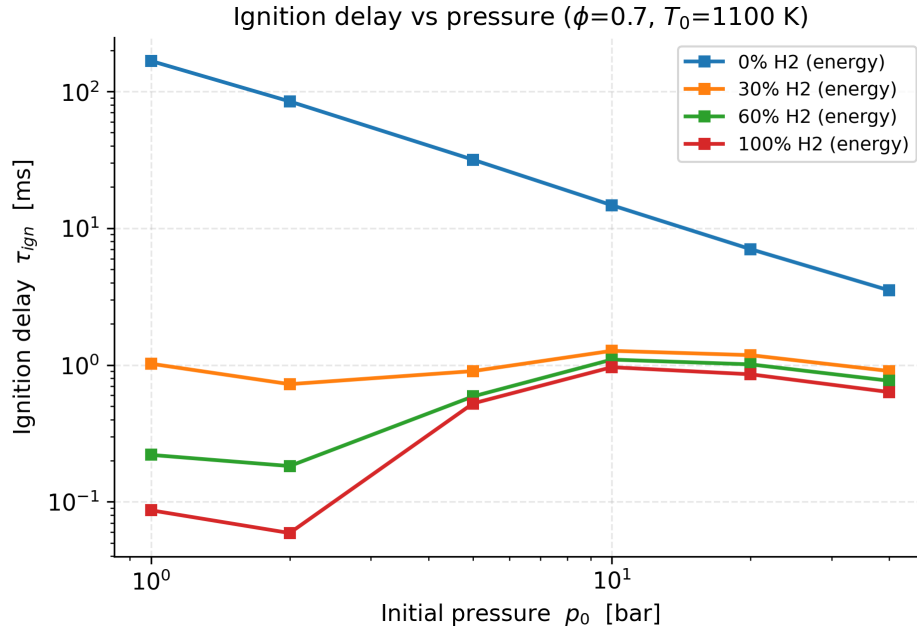


Figure 2: Ignition delay versus initial pressure for different hydrogen energy fractions ($\phi = 0.7$, $T_0 = 1100$ K).

$p = 20$ bar it falls from 7.0 ms for pure methane to 0.86 ms for pure hydrogen, with most of the reduction achieved by about 20 % hydrogen energy.

4.2 Laminar burning velocity

Figure 4 shows the laminar burning velocity as a function of equivalence ratio. For each blend the curve has the familiar shape with a maximum near or slightly above stoichiometric. Hydrogen has two effects: it raises the peak velocity dramatically and shifts the peak towards richer mixtures. The peak burning velocity rises from about 41 cm/s for pure methane to roughly 169 cm/s at 60 % hydrogen and well above 500 cm/s for pure hydrogen.

Figure 5 shows the burning velocity at stoichiometric conditions as a function of hydrogen fraction. The growth is strongly non-linear: S_L accelerates as the hydrogen content increases, reflecting the increasing dominance of the fast hydrogen chemistry and the higher flame temperature.

4.3 NO_x emissions

Figure 6 shows the NO_x emission index as a function of equivalence ratio. Each curve has a pronounced maximum at a slightly lean to near-stoichiometric mixture and falls off steeply on either side, the classical thermal-NO_x bell shape driven by the exponential temperature dependence of the Zeldovich route. Hydrogen shifts the whole family of curves upward: the peak emission index grows from about 101 g/kg for pure methane to over 460 g/kg for pure hydrogen.

The adiabatic flame temperature behind these emissions is shown in Figure 7. It rises with hydrogen content at every equivalence ratio, which is the proximate cause of the higher NO_x.

The central trade-off of hydrogen enrichment is made explicit in Figure 8, which plots the emission index and the flame temperature together against the hydrogen fraction at a fixed lean condition ($\phi = 0.7$). As the hydrogen energy fraction goes from 0 to 100 % the flame temperature rises

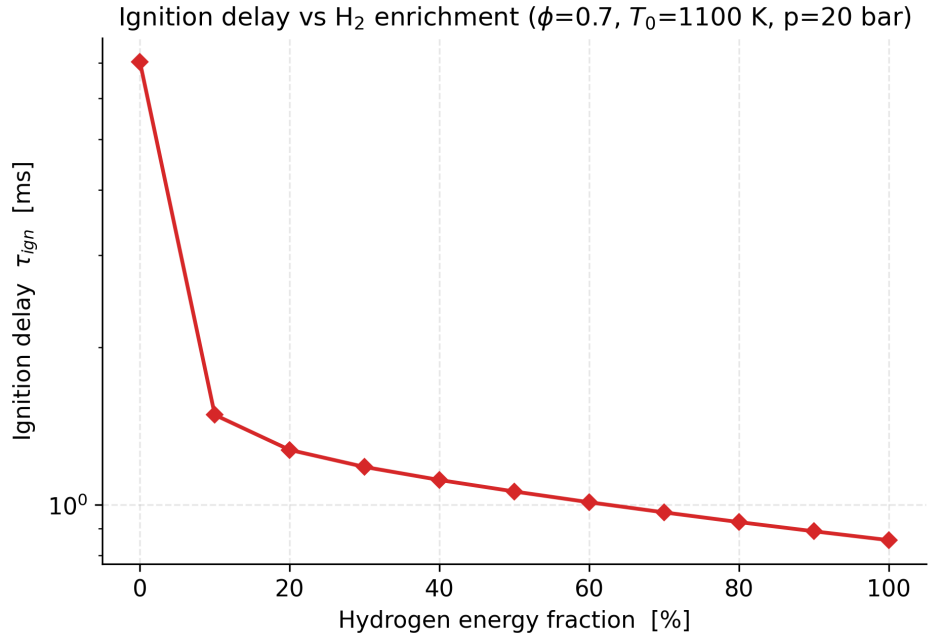


Figure 3: Ignition delay versus hydrogen energy fraction ($\phi = 0.7$, $T_0 = 1100$ K, $p = 20$ bar).

by only about 190 K, from 2139 K to 2327 K, while the emission index increases by more than a factor of twenty, from 8.8 g/kg to 185 g/kg. The flame temperature rises almost linearly, but the NO_x responds exponentially, which is exactly the expected signature of thermal NO_x .

4.4 Mechanism validation

Because GRI-Mech 3.0 is optimised mainly for natural gas, its reliability at the pure-hydrogen end of the blend range must be checked. The ignition delay and the laminar burning velocity of 100 % hydrogen were therefore recomputed with a dedicated detailed H_2/O_2 mechanism (10 species, 29 reactions) and overlaid on the GRI-Mech results in Figure 9. The agreement is excellent: the ignition delay differs by less than 0.1 % across the temperature range, and the laminar burning velocity by at most 3.2 %, with the largest deviation at the lean limit where the hydrogen chemistry is most sensitive. This confirms that GRI-Mech 3.0 is an appropriate single mechanism for the whole blend range.

5 Conclusions

The effect of hydrogen enrichment on methane–air combustion was computed with Cantera and GRI-Mech 3.0 over the full blend range, under conditions representative of an aero gas turbine combustor. The main findings are as follows.

Hydrogen substantially improves the ignition and burning characteristics. The ignition delay falls by almost an order of magnitude from pure methane to pure hydrogen, with most of the benefit obtained at modest hydrogen fractions, and the laminar burning velocity increases strongly and non-linearly, by roughly an order of magnitude at stoichiometric conditions. Both effects are favourable for relight, flame stabilisation and a compact reaction zone.

These benefits come at the cost of NO_x . Because hydrogen raises the adiabatic flame temperature and thermal NO_x depends exponentially on temperature, a modest temperature rise produces a

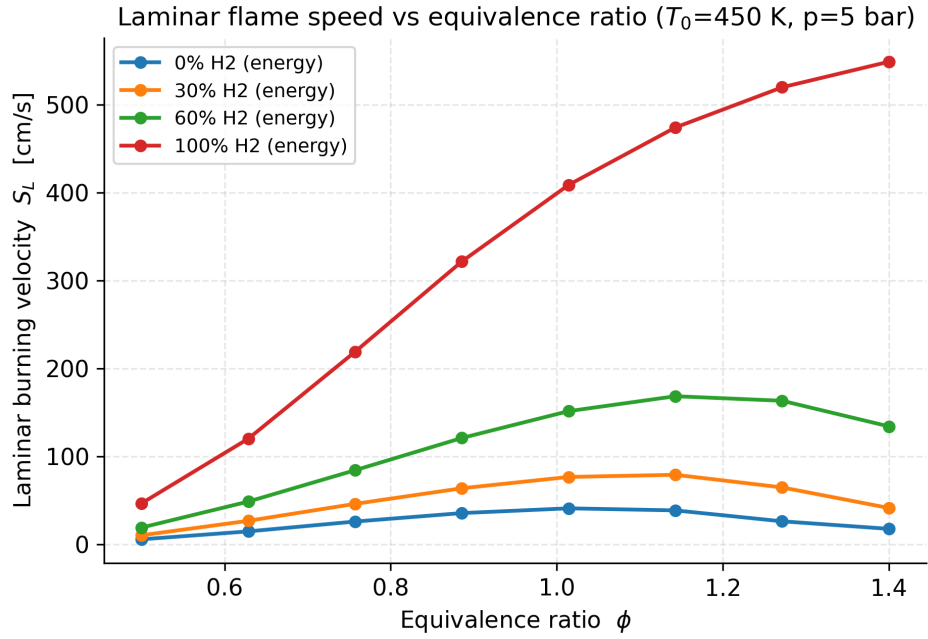


Figure 4: Laminar burning velocity versus equivalence ratio for different hydrogen energy fractions ($T_0 = 450$ K, $p = 5$ bar).

large emission penalty: at a lean condition the emission index increased more than twentyfold across the blend range for only a 190 K rise in flame temperature. Lean operation remains the principal means of keeping NO_x acceptable, and it becomes more important, not less, as the hydrogen fraction increases.

Finally, the use of a single mechanism across the whole blend range was justified: GRI-Mech 3.0 reproduces the predictions of a dedicated detailed hydrogen mechanism for both ignition delay and flame speed of pure hydrogen to within a few per cent. Overall, hydrogen enrichment is an effective route to decarbonising gas turbine combustion, but it shifts the design point firmly towards lean, carefully temperature-controlled operation in order to manage the NO_x trade-off.

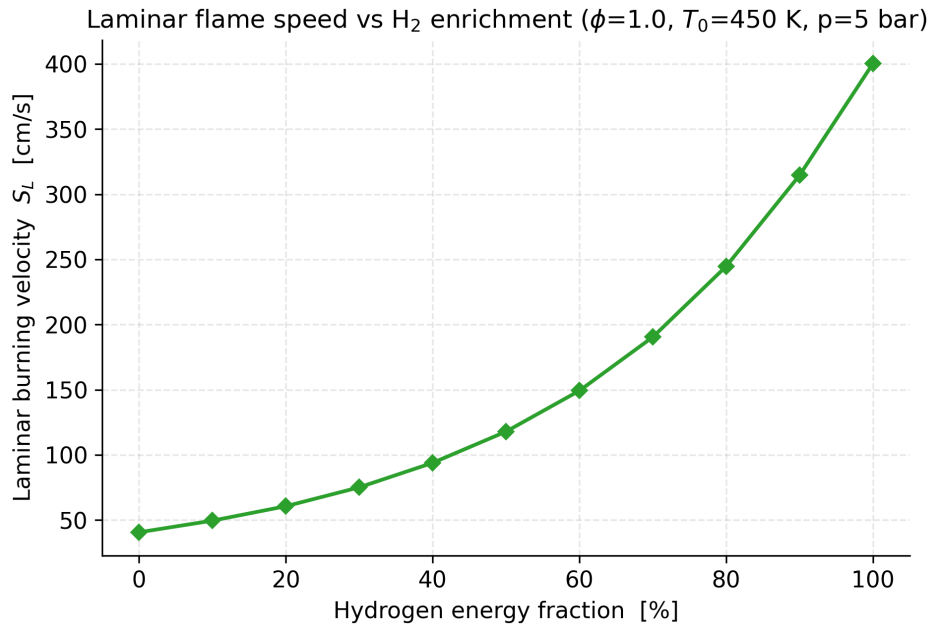


Figure 5: Laminar burning velocity versus hydrogen energy fraction at stoichiometric conditions ($\phi = 1.0$, $T_0 = 450$ K, $p = 5$ bar).

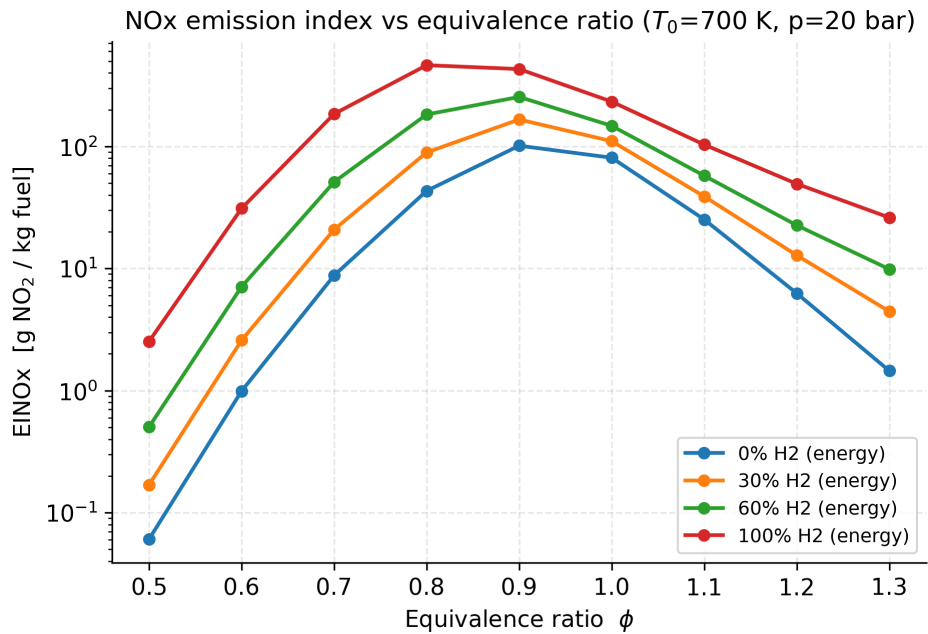


Figure 6: NO_x emission index versus equivalence ratio for different hydrogen energy fractions ($T_0 = 700$ K, $p = 20$ bar).

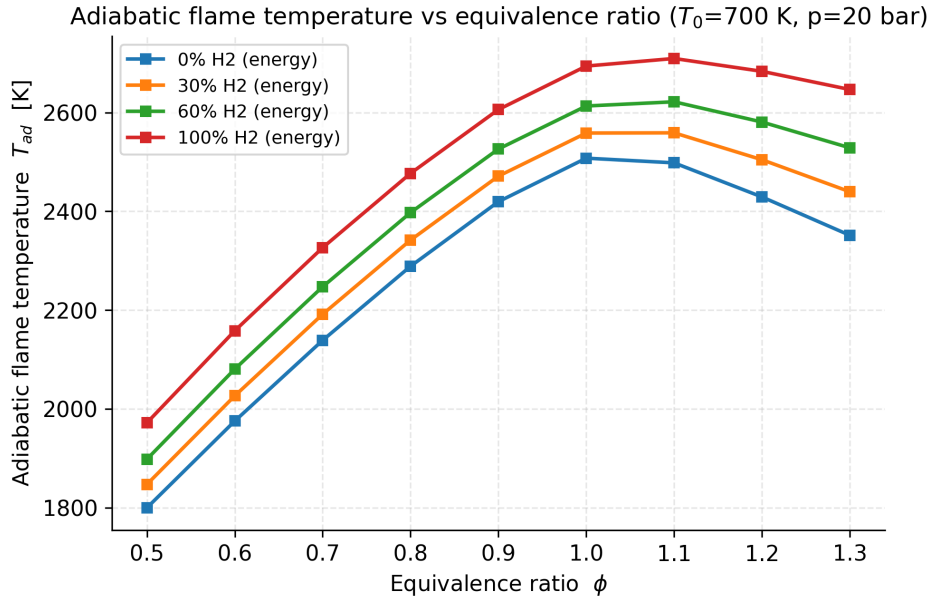


Figure 7: Adiabatic flame temperature versus equivalence ratio for different hydrogen energy fractions ($T_0 = 700$ K, $p = 20$ bar).

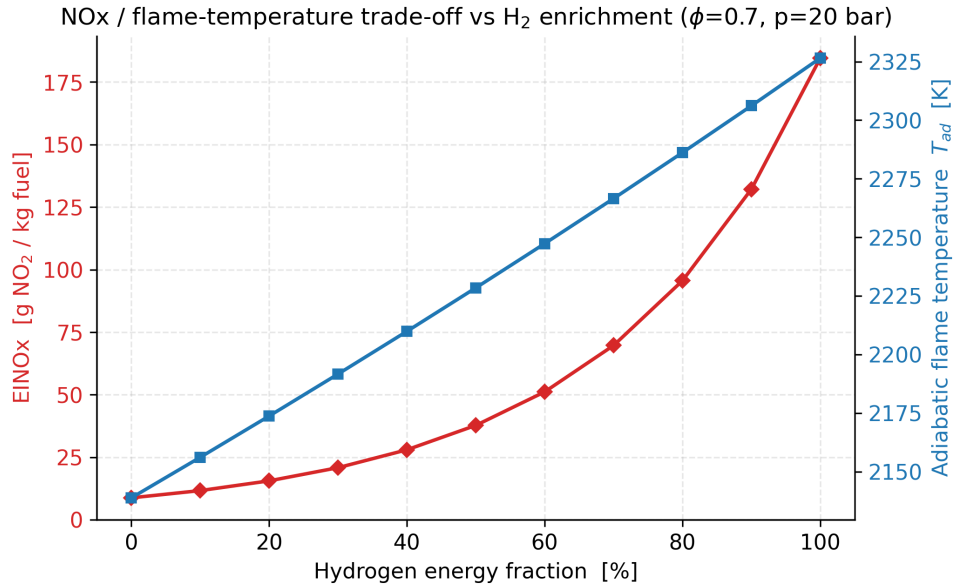


Figure 8: NO_x emission index and adiabatic flame temperature versus hydrogen energy fraction at a lean condition ($\phi = 0.7$, $p = 20$ bar).

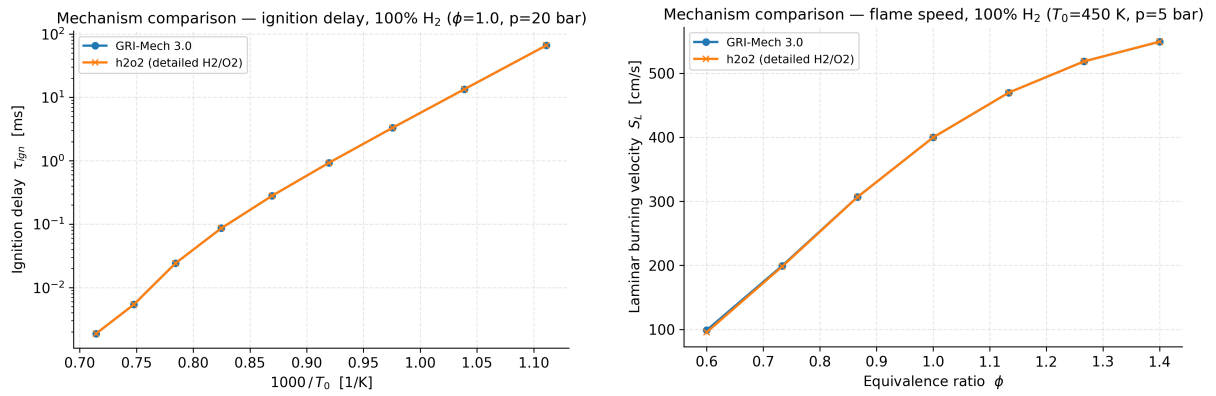


Figure 9: Mechanism validation for 100 % hydrogen: ignition delay (left) and laminar burning velocity (right), GRI-Mech 3.0 versus a detailed H₂/O₂ mechanism.